

ELEC 4700: Modelling of Integrated Device

Traveling Wave Model Project

Interim Project Report

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1 Introduction and Background

The main objective of this project is to use the Travelling Wave Model (TWM) to create a distributed laser model for optical wave propagation. The project, which is implemented in MATLAB, is divided into two primary phases: Passive Structures and Active Structures. Each phase is developed gradually through weekly milestones.

The goal of this extensive, multi-week numerical modelling assignment is to comprehend and describe the behaviour of light in optical waveguides. As the research progresses, different physical characteristics are investigated to see how they affect wave propagation, and numerical methods are gradually added to increase the simulation's precision and stability. The active phase will include carrier dynamics to replicate the functioning of optical amplifiers and lasers, while the passive phase focuses on simulating fundamental waveguide behaviour, gratings, and dispersion effects.

Ultimately, the project's goal is to provide a strong, adaptable framework that can facilitate in-depth research into intricate waveguide optics utilizing the Travelling Wave Model.

General Background Breakdown

The core of the project is built on the Travelling Wave Model, a set of partial differential equations that describe how forward- and backward-propagating optical fields evolve in time and space within a waveguide. The simulation work is implemented in MATLAB, using finite-difference time-domain (FDTD) and related numerical methods.

Key objectives include:

- Simulating optical signal propagation through waveguides.
- Introducing static gain/loss, dispersion, and grating-induced back-reflection.
- Modeling active devices like **optical amplifiers** and **lasers** by incorporating physical mechanisms such as **stimulated and spontaneous emission**.

Phase 2: Active Structures — Amplifiers and Lasers

The second phase of the project which will be the focus of the report focuses on active optical components, where carrier dynamics are integrated to simulate the behavior of optical amplifiers and lasers. This phase aims to enhance the Travelling Wave Model by incorporating key physical processes such as population dynamics and optical gain mechanisms. Core objectives include modeling carrier rate equations, simulating both stimulated and spontaneous emission, and analyzing quasi-steady-state conditions within active waveguides. Additionally, the effects of gain saturation and carrier depletion on pulse evolution will be studied in detail.

1. Carrier Equation – Milestone 6

A carrier rate equation is added to model carrier density dynamics $N(z)$ along the waveguide:

$$\frac{dN(z)}{dt} = \frac{\eta I_D}{qV_l} - G_o(N - N_{tr})S(z) - \frac{N(z)}{\tau_n}$$

Where:

- $\frac{\eta I_D}{qV_l}$: current injection (carrier generation)
- $G_o(N - N_{tr})S(z)$: stimulated emission term
- $\frac{N(z)}{\tau_n}$: spontaneous emission loss
- $S(z)$: photon density derived from optical field intensity

This equation is discretized and solved in the main simulation loop, updating the carrier distribution at each timestep, and allowing for carrier depletion modeling.

2. Optical Amplifiers and Spontaneous Emission – Milestone 7

$$\frac{1}{v_g} \frac{\partial E_f}{\partial t} = -\frac{\partial E_f}{\partial z} - i\beta(N, S)E_f + F_f$$

$$\beta(N) = \frac{i}{2}(g_f G_o(N - N_{tr}) - \alpha_l)$$

Where:

- $\beta(N)$: models gain and loss dynamically from the carrier density.
- F_f : spontaneous emission noise, implemented using random noise functions (e.g., randn in MATLAB) and scaled by a parameter SPE.

This allows the model to simulate real amplifier behavior, including gain saturation and noise buildup from spontaneous emission.

3. Lasers – Milestone 8

A laser cavity is simulated by introducing partially reflective mirrors and turning on spontaneous emission (SPE) as a noise seed:

- When the gain exceeds loss, the laser starts to oscillate, and steady-state laser output can be observed.
- The simulation captures both transient startup and mode evolution, analyzed in time and frequency domains.

- Adjusting parameters like drive current I_D , cavity length, and mirror reflectivity allows exploration of different laser behaviors.

The project concludes with an open-ended milestone where students can design and explore custom laser setups, such as DFB lasers, ring lasers, or lasers with external gratings or feedback mechanisms.

2 Carrier Equation – Milestone 6

Moving from a strictly passive waveguide model in the previous milestones 1 through 5 to an active structure that can amplify light is the aim of this milestone. To mimic stimulated emission, a crucial mechanism underlying both amplification and laser action, carrier dynamics are introduced into the simulation.

Physical Background

Charge carriers, such as electrons and holes, are injected into the active medium of semiconductor optical devices, such as SOAs and lasers, by means of an external current.

Three main routes exist for these carriers to recombine:

- Stimulated Emission: Coherent photons are released when an optical field is present, which causes carrier recombination and optical gain.
- Noise is a result of carriers recombining at random to produce photons that are not aligned with the input wave.
- Carrier Lifetime Decay: Even in the absence of optical stimulus, carriers naturally recombine with time.

Finally, a carrier rate equation is presented to quantitatively simulate this behaviour.

The Carrier Equation

Recalling the carrier equation from the introduction where the carrier density $[N(z, t)]$ varies over time and space is governed by:

$$\frac{dN(z)}{dt} = \underbrace{\frac{\eta I_D}{qV_l}}_{\text{Injection (source)}} - \underbrace{G_0(N(z) - N_{tr})S(z)}_{\text{Stimulated emission (gain)}} - \underbrace{\frac{N(z)}{\tau_n}}_{\text{Spontaneous decay}}$$

Figure 1: The carrier rate equation

Where:

$$\frac{\eta I_D}{qV_l}:$$

- carrier injection from the driving current I_D
- η ; the quantum efficiency
- q : the electron charge
- V_l : the volume of the active region

$$G_o(N - N_{tr})S(z):$$

- The stimulated emission term
- G_o ; the gain coefficient
- N_{tr} ; transparency carrier density which is the amplification threshold
- $S(z)$; the photon density, calculated from the field intensities

The field intensities are calculated using this equation:

$$S(z) = CE_{to}P (|\hat{E}_f(z)|^2 + |\hat{E}_r(z)|^2)$$

Where the $CE_{to}P$ is the conversion factor from field strength to photon density.

$$\frac{N(z)}{\tau_n}:$$

- Models the spontaneous emission loss over the carrier lifetime.

Overall, because of its ease of use and compatibility with time-domain simulations, the Euler approach was employed for discretization. It guarantees computational efficiency and offers a clear view of each term's physical contributions.

MATLAB Implementation

The MATLAB code developed for this milestone follows a structure similar to previous ones. It begins by initializing all relevant physical constants and system parameters, including the speed of light, electron charge, and the permittivity of free space. For this milestone, additional parameters such as carrier lifetime, transparency carrier density, and the dimensions of the active region are also defined. A spatial grid is created to discretize the laser cavity length, and the initial carrier density is set uniformly across this grid—typically near the transparency carrier density—to ensure realistic starting conditions.

A Gaussian pulse stream is generated and introduced into the system. An electrical bias current is used to continually inject carriers into the active region while this optical input interacts with the carriers in the medium. To observe the carrier response under different biasing conditions, the simulation alternates between a strong injection current during the pulse stream and a reduced background current outside the pulse window.

The photon density influences the emission process by depleting the carriers when light interacts with the medium. There is a total change in the carrier density which is determined by balancing the carrier injection from the applied current with the carrier depletion and other losses.

The carrier density is updated at each time step using the Euler integration method, which offers enough precision to follow the dynamics of fast carrier modulation brought on by the laser pulses. Although understanding carrier behaviour is the focus here, the propagation of the optical field within the cavity is then influenced by the changing carrier population.

The differential equation is discretized and integrated into the main loop of the simulation. A simple forward Euler update might be seen in the image or Figure below.

```
% Modified and updated carrier density
Stim = gain.*(N - Ntr).*S;
N = (N + dt*(I_injv/ eVol - Stim))./(1 + dt / taun);
N(N < 0) = 0;
Nave(i) = mean(N);
```

Code Snippet 1: The code snippet showing the simple forward Euler update

- **dt** is the time step
- **N** is updated at every step using the latest values of **S(z)**, the photon density.
- The model also tracks the carrier density along the waveguide [**N(z)**]
- **Nave(i)** is the average carrier density over time which becomes useful for monitoring the system behaviour like a gain saturation.

In a simulated setting, the implementation successfully conveys the fundamental physics of the carrier rate equation. This MATLAB code offers insight into the dynamical behaviours of this system and equation and enables the study of how carriers react to laser excitation.

When running the simulations under different conditions, there are things or scenarios that can be visualized which include:

N(z) and S(z) profiles:

- Show how carriers are depleted in regions of strong light intensity.
- Reveal gain clamping in amplifier regions.

Carrier response to pulsed current:

- Simulates how the carrier density rises/falls based on IDID
- Pulsed drive can model modulation in real laser diodes.

Carrier depletion from input optical pulses:

- A Gaussian pulse will “consume” carriers via stimulated emission as it propagates.

Finally, the carrier equation is fundamental for simulating realistic optical amplifiers and lasers. If this equation is not present, the model lacks the non-linear dynamics needed to represent the gain saturation, noise dynamics, and the lasing thresholds. Furthermore, the carrier dynamics enable the simulation to be self-consistent; the light affects carriers and vice-versa, as the case is in real photonic devices.

Response to pulsed ID with optical sources off

When the optical sources are shut off, the forward electric field is close to zero. Thus, there are no optical sources causing stimulated emission, the position on the carrier density vs. position plot is flat. Finally, the average carrier density naturally decays after first rising because of the current injection.

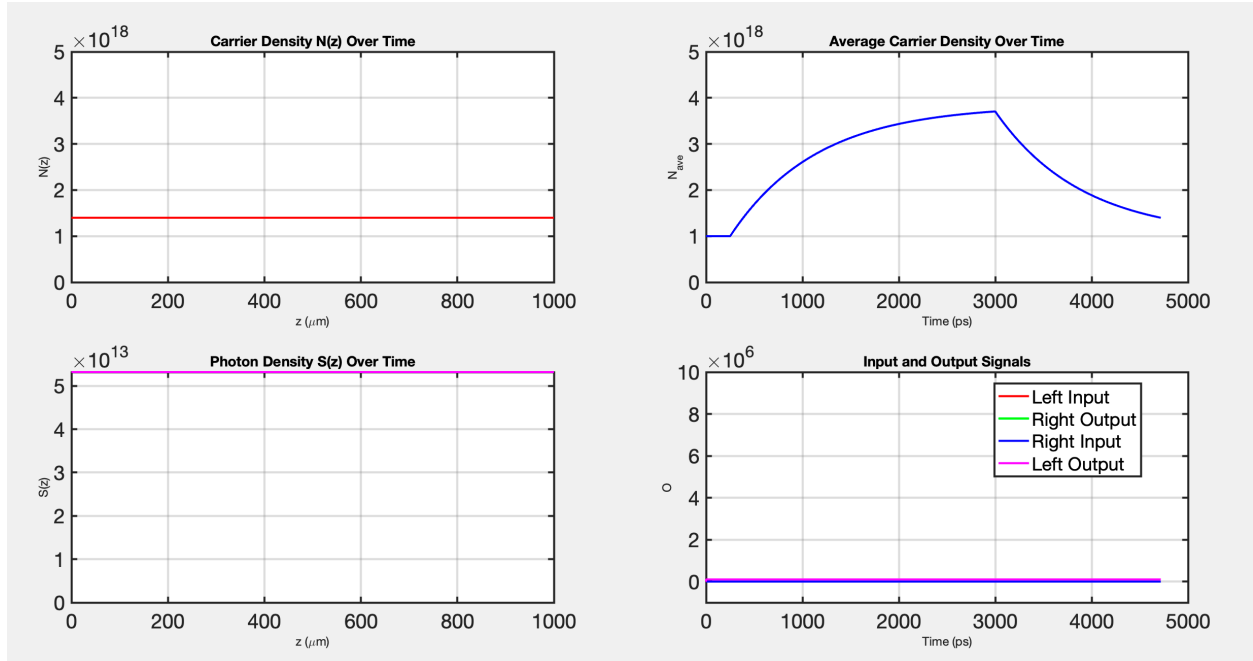


Figure 2: The four plots required for the milestone; they show the responses to pulsed ID with the optical sources turned off

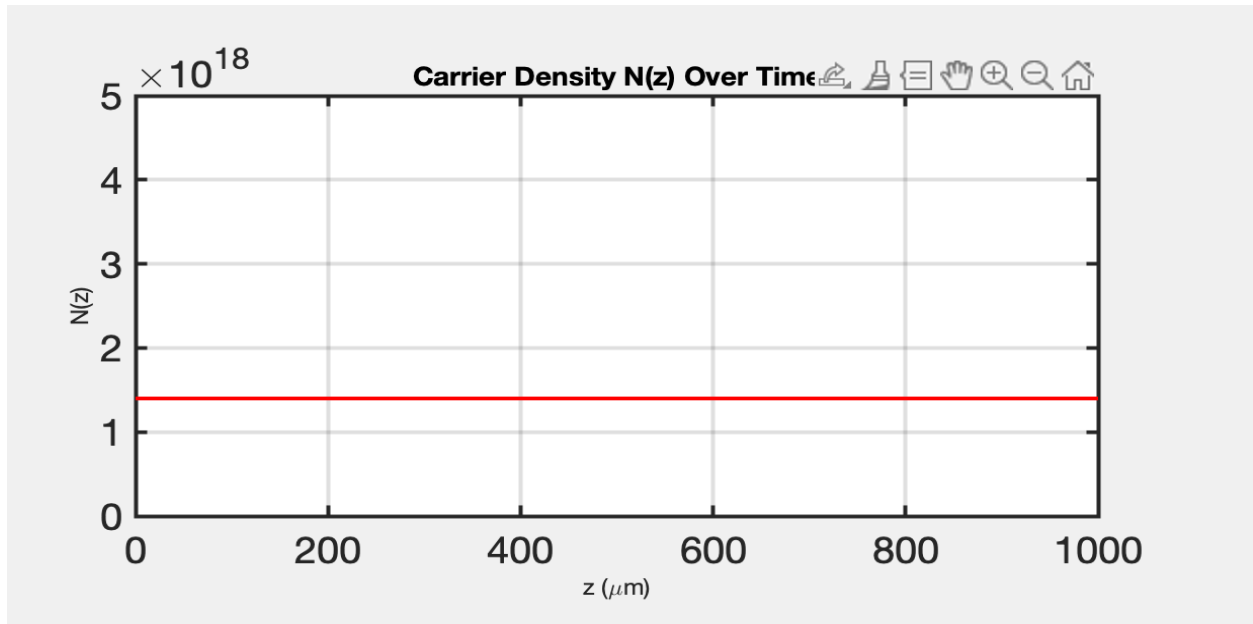


Figure 3: The plot of the carrier density over time when the optical sources are turned off

The plot above in Figure 3 is flat due to absence of propagation of optical fields which helps to stimulate emissions spatially.

Furthermore, in **Figure 4** below there is no stimulated emission to balance generation. Hence, the plot (carrier density) rises initially before decreasing (decaying) due to the domination of recombination.

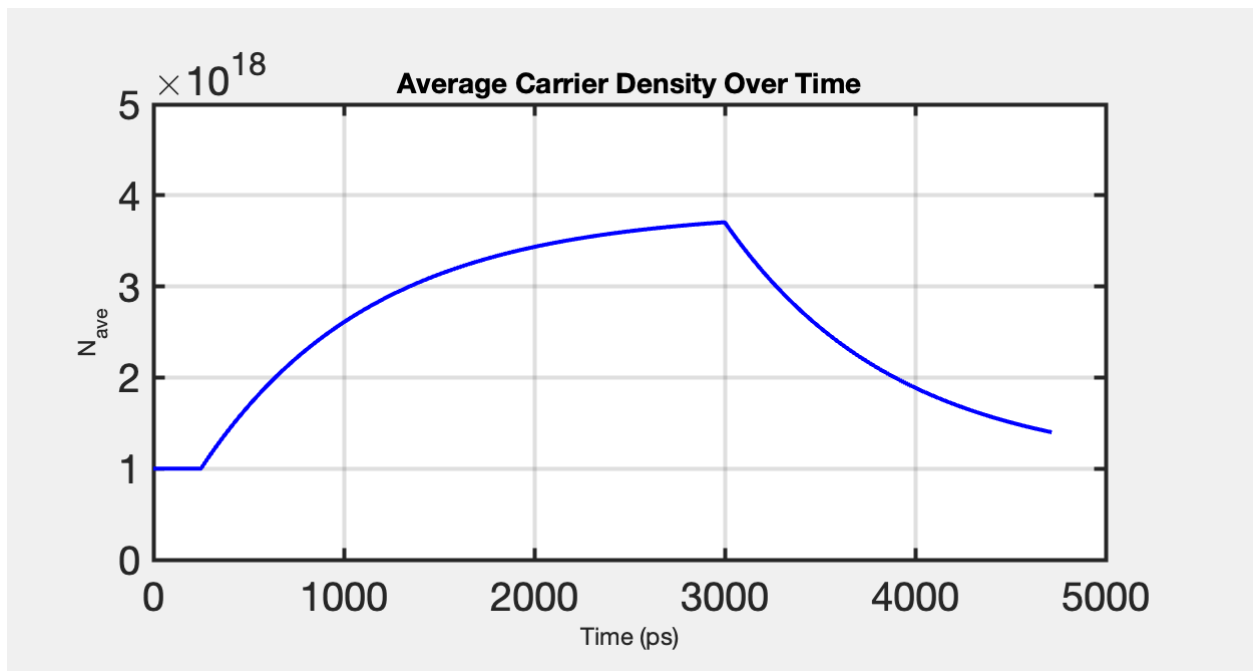


Figure 4: The plot of average carrier density over time when the optical sources are turned off

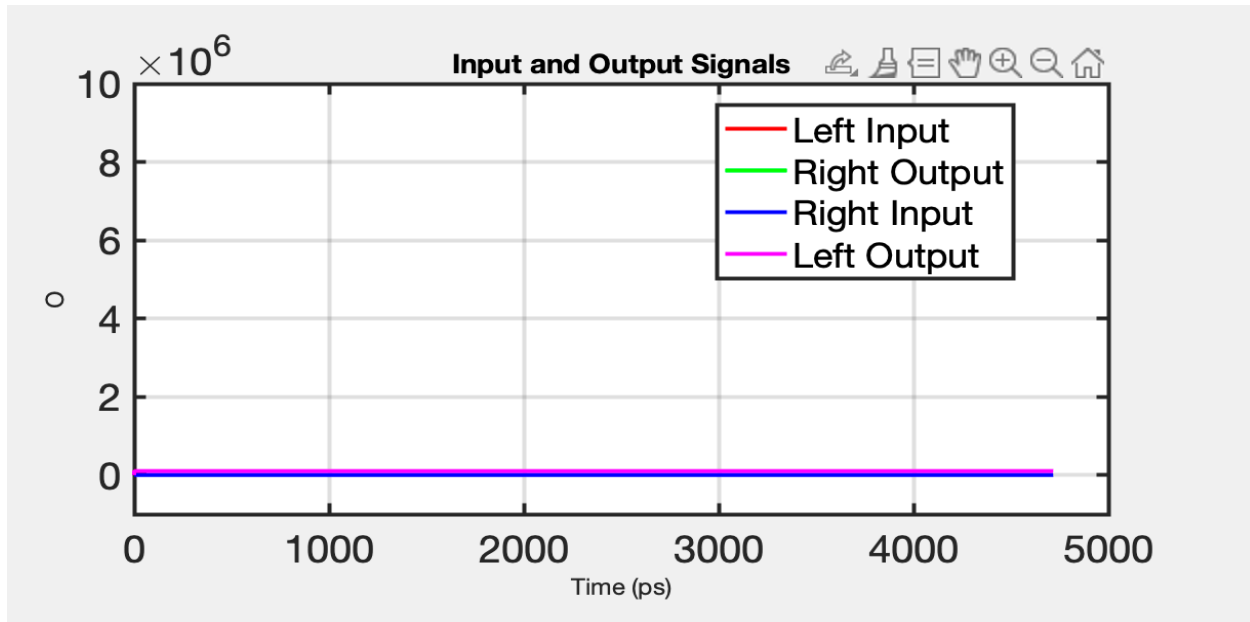


Figure 5: The plot of the input and output signals when the optical sources are off (these are already set to 0)

Response to a gaussian pulse stream

The optical sources are controlled by the values computed in the line of code provided below:

```

InputL(i) = SourceFct_M6(t, InputParasL);

% Input pulse parameters (Modulated Gaussian)
InputParasL.E0 = 9e5; % Amplitude (adjusted to match the scale in the image, e.g.,  $\times 10^6$ )
InputParasL.we = 200e-12; % Angular Frequency
InputParasL.t0 = 10e-12; % Time offset (in seconds, adjusted for ps scale)
InputParasL.wg = 1e-12; % Pulse width (in seconds, adjusted for ps scale)
InputParasL.phi = 0; % phase
InputParasL.type = 'sinc'; % Initialize the new pulse type
InputParasL.rep = 500e-12;
InputParasR = 0; % Right input

```

Code Snippet 2: The code snippet showing what controls the optical source (either ON/ OFF)

The “InputL(i) = SourceFct_M6(t, InputParasL)” line calls the function “SourceFct_M6”, which generates the optical input field at each time step based on the pulse parameters defined earlier (also in the Figure above). The implementation of “SourceFct_M6” determines the pulse's shape and timing (i.e., when it is turned "on" or "off").

```

function E = SourceFct_M6(t,InputParas)

if isfield(InputParas, 'rep')
    n = floor(t/InputParas.rep);
    t = t-n * InputParas.rep;
end

if ~isstruct(InputParas)
    E = InputParas;
else
    if isfield(InputParas, 'type') && strcmp(InputParas.type, 'sinc')
        % New Sinc Pulse: E0 for t0 < t < (t0 + wg), otherwise 0
        E = InputParas.E0 * (t >= InputParas.t0 & t <= (InputParas.t0 + InputParas.wg));
    else
        E = InputParas.E0*exp(-(t-InputParas.t0)^2/InputParas.wg^2)*exp(1i*(InputParas.wg*t + InputParas.phi));
    end
end
end

```

Code Snippet 3: The code snippet showing the implementation of the function SourceFct_M6

In the code snippet above (Figure 8), the function uses various variables or terms to generate a pulse train (pulse shape) and switch the optical source on (during specific pulse windows) or off (zero outside the pulse window) during specific pulse windows.

When the optical source is “ON”, the current time is within the width of one of the optical pulses generated by the function in the code snippet provided in Code Snippet 3 as well as the value of the amplitude is adjusted from “0” to “ $9e5$ ”. Whereas, when the optical source is turned “OFF” the t is outside the active region of the pulses. Furthermore, the responses to the optical sources turned off have been shown or highlighted in previous figures above. Nevertheless, the response to the optical sources turns “ON” can be highlighted in the figures below.

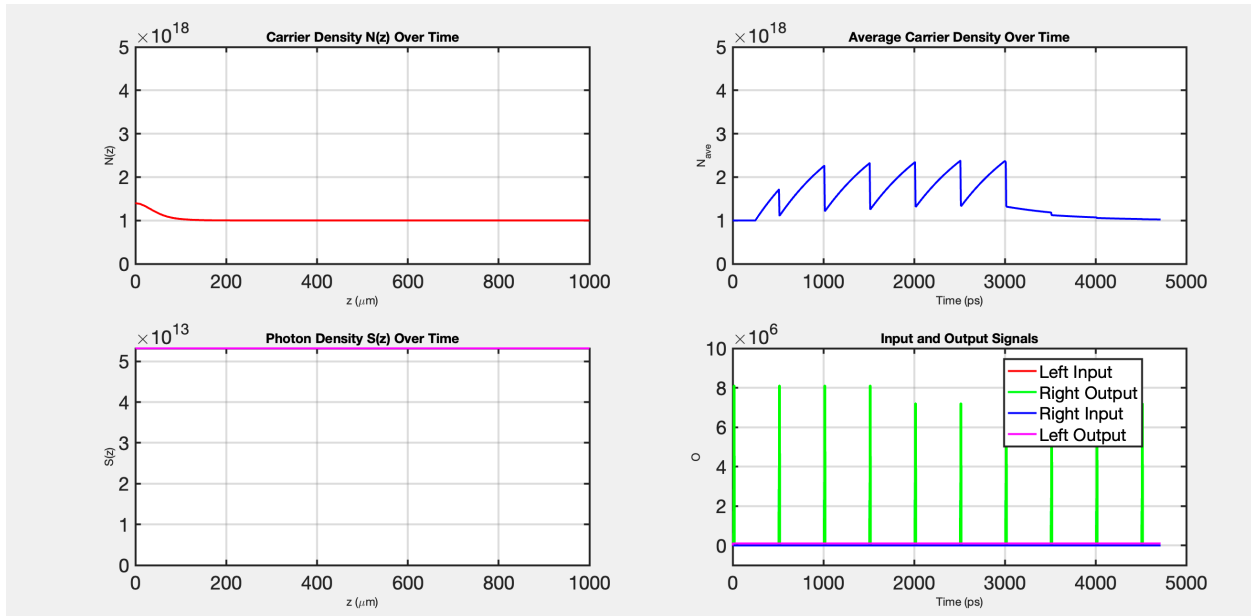


Figure 6: The four plots required for the milestone; they show the response to pulse I_D with the optical sources turned ON

This plot is like the one shown when the optical sources was turned “OFF”. However, there is a Gaussian pulse stream that changes the outputs of the system significantly.

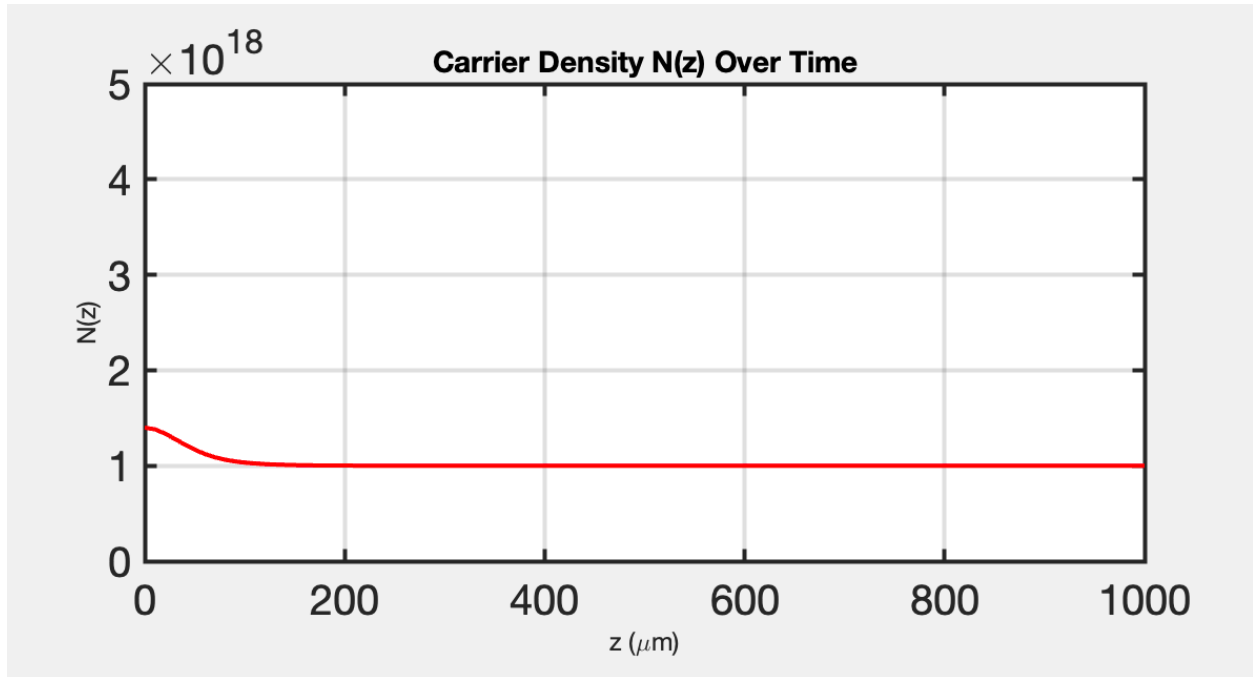


Figure 7: The plot of carrier density over time when the optical sources are turned ON

A note on Figure 7, although the carrier distribution in this figure is mostly flat, it should be noted that there are minor changes that arise at specific locations along the cavity where the optical pulse causes carrier depletion.

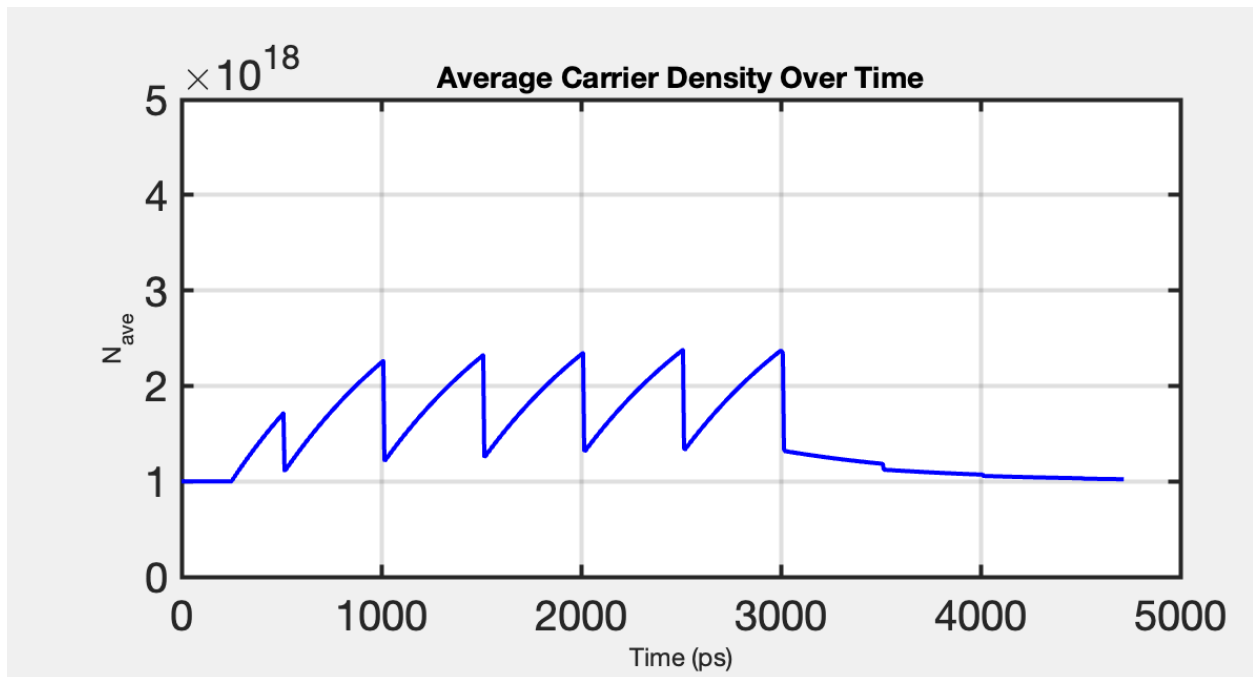


Figure 8: The plot of the average carrier density over time with the optical sources turned ON

The carrier density in this plot above (Figure 8) clearly exhibits oscillatory behavior, spiking and dipping frequently, which is directly related to the injected Gaussian pulse stream's periodic nature. The dips show the carrier depletion brought on by the stimulated emissions when the pulse arrives, while the peaks show the carrier accretion during the idle periods.

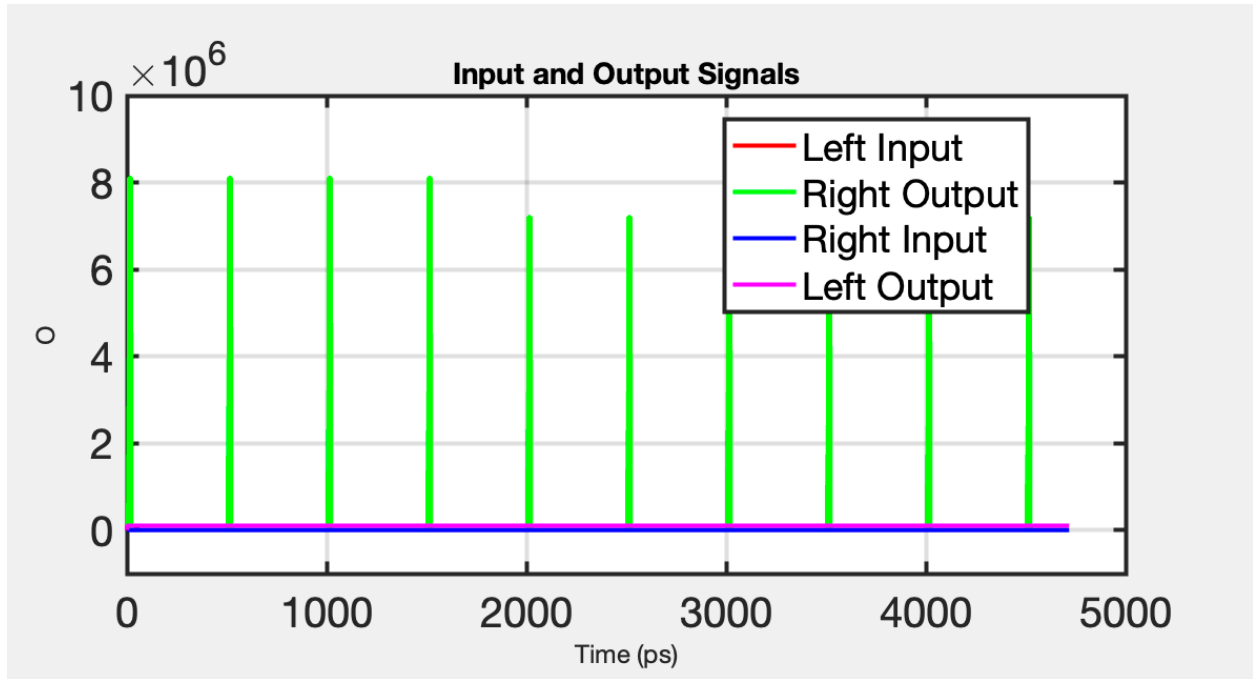


Figure 9: The plot of the inputs and outputs generated when the optical sources are turned ON

Finally in Figure 9 above, the input and output signals from the optical pulse streams is displayed in the plot. Periodic peaks in the plot attest to the Gaussian pulse stream's successful optical output and transmissions throughout the active medium.

The impact of both electrical and optical injection on carrier dynamics in the semiconductor medium is evident from the simulation results from Milestone 6. Applying a pulsed electrical injection when the optical sources are off causes the carrier density to increase because of electrical pumping. As recombination balances the injection rate, the carrier density eventually saturates. The optical field and output signal are insignificant during this stage. When a Gaussian pulse stream is introduced, observable fluctuations in the mean carrier density appear over time, which correspond to the periodic loss and re-supply of carriers in response to each optical pulse. Along the device's length, the carrier distribution that results shows a more dynamic profile. These results highlight the anticipated interplay between electrical pumping and optical excitation and confirm the correctness of the applied carrier rate equation.

3 Optical Amplifiers and SPE – Milestone 7

The goal of this milestone is to put into practice a thorough optical amplifier model that takes into consideration spontaneous emission noise (SPE), which adds randomness to the system, as well as stimulated emission, the main amplification mechanism. Understanding the behaviour of

semiconductor optical amplifiers (SOAs) and lasers requires an understanding of these two basic processes.

Because they immediately increase signal intensity without requiring optical-to-electrical conversion, optical amplifiers are essential components of optical communication systems. This increases the overall quality of the signal and increases the transmission distance. The simulation increases the realism of active device modelling by expanding on the carrier dynamics shown in Milestone 6.

Scenarios Explored

The amplifier's behavior is evaluated under four distinct conditions:

1. Pure amplification of a pulse stream without noise
2. Amplification of spontaneous emission noise (SPE) in the absence of an input pulse, with gain polarization
3. Amplification of SPE without gain polarization and no input source
4. Influence of injection current (I_D) and input pulse configuration on amplifier performance

Signal amplification in a semiconductor optical amplifier is accomplished through stimulated emission, in which incident photons cause excited electron-hole pairs to recombine, emitting identical photons and coherently boosting the optical field.

Nevertheless, noise is introduced by spontaneous emission, in which carriers recombine at random to release photons with uncorrelated phase and direction. This process is crucial in lasers, where it "seeds" the optical field that becomes coherent by stimulated emission, but it is seen as undesirable in amplifiers.

The model uses a connected collection of physical models to realistically replicate SOAs by incorporating both mechanisms:

- The Carrier Rate Equation explains how carriers inject, recombine, and modulate over time.
- Modified to incorporate complicated gain reliant on carrier density, travelling wave equations model both forward and backward propagating optical fields.
- The Stochastic SPE Source Term simulates noise effects by adding unpredictability to the optical field.

These elements work together to create a realistic simulation framework that can be used to study noise behaviour and signal amplification in the frequency and temporal domains.

$$\beta(N) = \frac{i}{2}(g_f G_o(N - N_{tr}) - \alpha_l)$$

The stimulated emission term ($\beta(N)$) models both the optical gain and waveguide loss.

- g_f : field gain factor (empirical tuning parameter)
- G_o : gain coefficient (related to cross-section and material properties)
- N_{tr} : transparency carrier density (minimum needed for net gain)
- α_l : The optical loss (absorption and scattering)

The stimulated emission term causes exponential amplification or attenuation of the field depending on the sign of $N - N_{tr}$

The spontaneous emission term: $F_f F_r$ represents the random field perturbations that simulate the spontaneous emission noise. It is implemented using the complex Gaussian random variables:

$$F_f = SPE (randn + i randn)$$

The “randn” is a MATLAB function generating normally distributed (Gaussian) random numbers.

Furthermore, the SPE is a scalar parameter controlling the intensity of spontaneous emission, when it has a value of ‘0’, no noise is introduced. Thus, when it is increased from that value of zero, it acts as a realistic amplifier noise and enables the laser to startup.

This stimulated emission term is acquired from the equation used in milestone 6, the carrier rate equation which was changed from:

$$\frac{dN(z)}{dt} = \frac{\eta I_D}{qV_l} - G_o(N - N_{tr})S(z) - \frac{N(z)}{\tau_n}$$

to:

$$\frac{dN}{dt} = \frac{\eta I_D}{qV} - G(N)S - \frac{N}{\tau_n}$$

due to the dependence of the $G(N)$ on the carrier density. This equation was discretized into MATLAB using the Euler method.

To complete the MATLAB simulation, the next essential component was the travelling wave equation, which governs the evolution of the optical fields within the amplifier. This equation accounts for both the forward-propagating field (E_f) and the backward-propagating field (E_r). It incorporates several key physical effects, including group velocity, spatial propagation, complex gain/loss, and any coupling mechanisms.

The general form of the travelling wave equations is given by:

$$\frac{1}{v_g} \frac{\partial E_f}{\partial t} = -\frac{\partial E_f}{\partial z} - i\beta(N)E_f + F_f$$

$$\frac{1}{v_g} \frac{\partial E_r}{\partial t} = -\frac{\partial E_r}{\partial z} - i\beta(N)E_r + F_r$$

Where:

- V_g is the group velocity
- $\beta(N) = \frac{i}{2}(G(N) - \alpha)$ represents the complex propagation constant, dependent on carrier density N
- $G(N)$ is the gain, while α is the loss coefficient
- F_f and F_r are the stochastic noise terms modeling spontaneous emission for forward and backward fields, respectively.

$$\begin{array}{l} \frac{1}{v_g} \frac{\partial \hat{E}_f}{\partial t} = -\frac{\partial \hat{E}_f}{\partial z} - \underbrace{i\hat{\beta}(N, S)\hat{E}_f}_{\text{Stimulated Emission}} + \underbrace{\hat{F}_f}_{\text{Spontaneous Emission}} \\ \frac{1}{v_g} \frac{\partial \hat{E}_r}{\partial t} = +\frac{\partial \hat{E}_r}{\partial z} - \underbrace{i\hat{\beta}(N, S)\hat{E}_r}_{\text{Stimulated Emission}} + \underbrace{\hat{F}_r}_{\text{Spontaneous Emission}} \end{array}$$

Figure 10: The screenshot of the equation above showing the components or terms which refer to the stimulated and spontaneous emission

The inclusion of spontaneous emission (SPE) as a stochastic noise process is a crucial component of the model. The simulation uses random complex noise, which is created by generating white Gaussian noise using MATLAB's randn function. To make sure that the noise power accurately represents the spontaneous emission rate, which is a function of the carrier density, a scaling factor A is added. As a result, both forward and backward noise generation can be realistically simulated by the model.

These equations are implemented in MATLAB through the following code snippets which will be provided as a screenshot or image(s) below.

```

S = (abs(Ef).^2 + abs(Er).^2).*EtoP*1e-6;
if t < Ion || t > Ioff
    I_injv = I_off;
else
    I_injv = I_on;
end
Stim = gain.*(N-Ntr).*S;
N = (N + dt*(I_injv/eVol - Stim))./(1+ dt/taun);
Nave(i) = mean(N(:));

```

Code Snippet 4: The code implementing the carrier rate equation

```

% Update forward and backward fields with grating terms
Ef(2:Nz) = fsync*exp_det(1:Nz-1).*Ef(1:Nz-1) + 1i * dz * kappa(2:Nz) .* Er(2:Nz);
Er(1:Nz-1) = fsync*exp_det(2:Nz).*Er(2:Nz) + 1i * dz * kappa(1:Nz-1) .* Ef(1:Nz-1);

Ef(2:Nz-1) = Ef(2:Nz-1) - LGain * (Ef(2:Nz-1)- Pf(2:Nz-1));
Er(2:Nz-1) = Er(2:Nz-1) - LGain * (Er(2:Nz-1)- Pr(2:Nz-1));

```

Code Snippet 5: The code implementing the forward and backward wave propagation

```

A = sqrt(gamma*beta_spe*c_hb*f0*L*1e-2/taun)/(2*Nz);
if SPE > 0
    eTf = (randn(Nz,1)+1i*randn(Nz,1))*A;
    eTr = (randn(Nz,1)+1i*randn(Nz,1))*A;
else
    eTf = (ones(Nz,1))*A;
    eTr = (ones(Nz,1))*A;
end

```

Code Snippet 6: The code implementing the introduction of noise

The MATLAB simulation in Milestone 7 extends the optical system model to incorporate both stimulated and spontaneous emission processes within a semiconductor optical amplifier (SOA), building on the framework created in Milestone 6. More accurate simulations of active photonic devices under varied operating conditions are now possible because to this improvement.

Physical constants and system parameters, such as carrier lifetime, gain coefficients, spontaneous emission (SPE) factors, and transparency carrier density, are initialized at the start of the simulation. The carrier density is initialized at the transparency level to allow for significant interaction with injected optical fields, and a spatial grid is created to discretize the amplifier throughout its length.

To mimic the optical input, a stream of Gaussian pulses is supplied at one end of the amplifier. The medium interacts with this propagating pulse, and carriers are continuously supplied by electrical injection current. This current is dynamically modulated by the simulation: a lower bias between pulses aids carrier recovery, while a greater bias is applied during the optical pulse to encourage gain. This modulation makes it possible to examine the ways in which electrical bias patterns affect signal amplification and carrier dynamics.

Photon density is computed from the intensities of the forward and backward fields and is used to model stimulated carrier depletion. The overall change in carrier density is calculated by balancing three key processes:

1. Carrier injection from the applied current
2. Carrier depletion via stimulated emission
3. Natural recombination losses

The Euler method is used for iterative time-domain updates, providing sufficient resolution to capture rapid carrier modulation due to electrical and optical interactions.

A key advancement in this milestone is the inclusion of spontaneous emission noise. At each simulation step, random complex noise is generated using `randn`, scaled by physical constants such as photon energy and spontaneous emission factors. These noise terms are injected into both E_f and E_r , simulating Amplified Spontaneous Emission (ASE), which is especially important when analyzing amplifier behavior in the absence of an input signal.

Additionally, gain polarization effects are modeled, capturing the amplifier's material response to the optical field, including gain saturation and spectrum broadening. These effects, in combination with dynamic carrier density, significantly influence the amplification process.

Throughout the simulation, key data—including input/output fields, carrier density, and SPE contributions—are continuously recorded. Real-time plotting visualizes the system's evolution, offering insights into the temporal behavior of the amplifier.

As a final step, a Fast Fourier Transform (FFT) is applied to the output fields, providing frequency-domain analysis of the amplified signals. This allows for an examination of spectral broadening, phase response, and overall signal integrity.

Finally, this milestone allows us to create a realistic MATLAB-based model of an optical amplifier that incorporates spontaneous emission noise for noise modelling and stimulated emission for signal gain is successfully implemented. It provides time and frequency domain insights into amplifier behaviour under a variety of operating settings by simulating the complex interactions between carrier dynamics, optical field propagation, and electrical injection control. The simulation provides a strong basis for future research into noise-dominated regimes and laser dynamics.

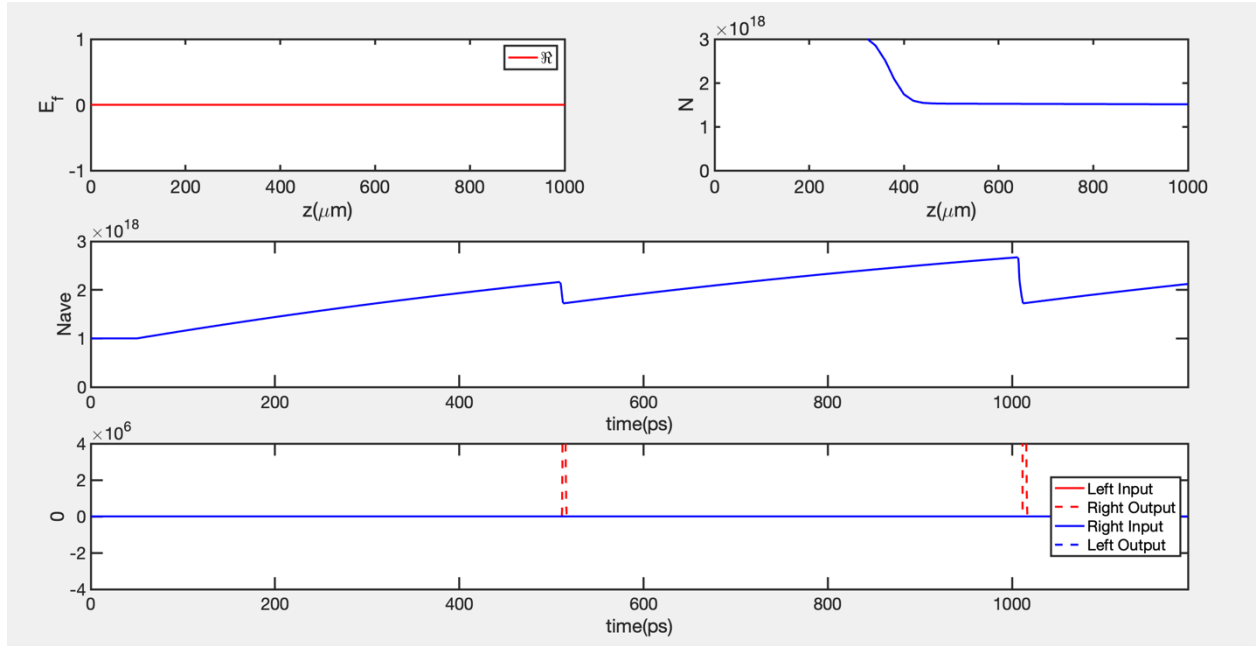


Figure 11: Time Domain evolution of the optical signal and carrier density without SPE

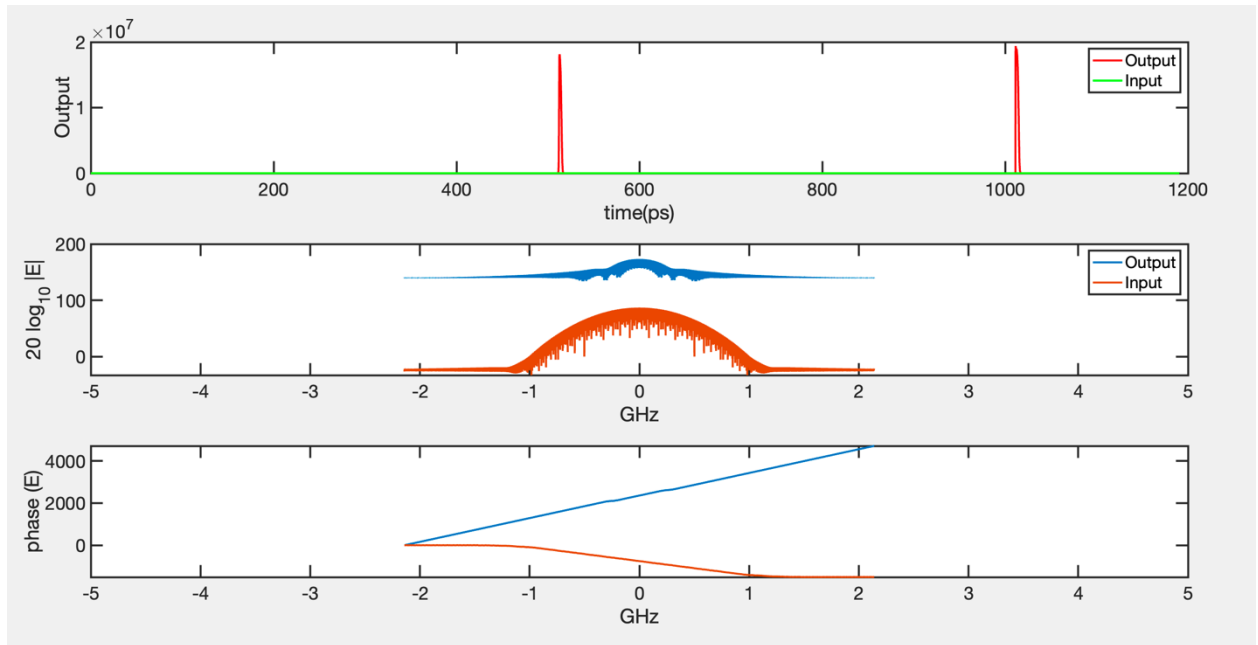


Figure 12: The frequency domain evolution of the optical signal and carrier density without SPE

In this case, an input pulse stream is used to study the system, and spontaneous emission is excluded. With clearly defined input pulses and a noticeably increased output, the time-domain plot validates efficient signal amplification. In accordance with expected behaviour, the carrier density map shows a regular pattern in which carriers are reduced during each pulse and rebound in the intervals between them. Furthermore, the output spectrum in the frequency domain displays observable broadening—a consequence of the amplification process—while the phase

plot clearly shows modulation, demonstrating appropriate signal propagation through the gain medium.

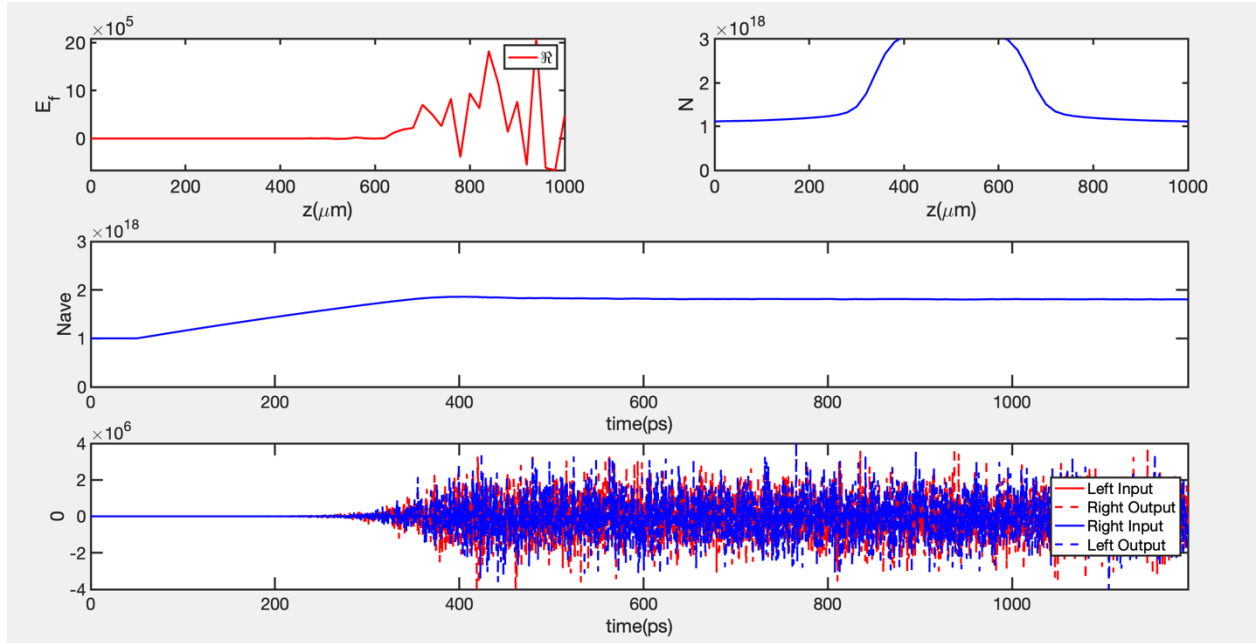


Figure 13: The time domain wave propagation of the optical signal and carrier density with SPE and gain polarization

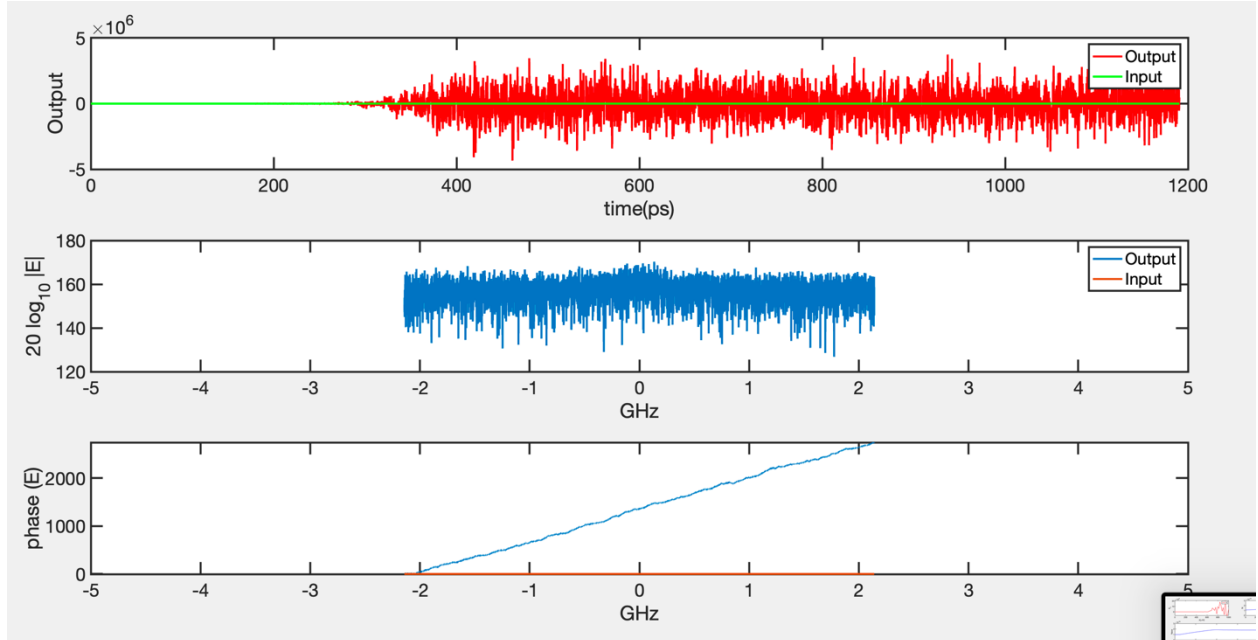


Figure 14: The frequency domain of the wave propagation of the optical signal and carrier density with SPE and gain polarization

In this case, the system is simulated with active gain polarization and without any external input source, depending only on spontaneous emission. The carrier density rises close to the centre of

the active zone because of the continuous gain, but the forward electric field stays at the noise level, as seen in Figure 13. The average carrier density stabilizes at a greater level when there is no intense optical input to deplete the carriers. The gain medium's amplification of spontaneous emission noise is reflected in the output signal.

The frequency-domain output is shown in the figure 14, where the spectrum seems widened and the time-domain trace shows oscillations typical of spontaneous emission, which is consistent with the emission's random and incoherent nature. This randomness is further demonstrated by the phase plot. All these findings are consistent with how spontaneous emission amplification in an active optical medium should behave.

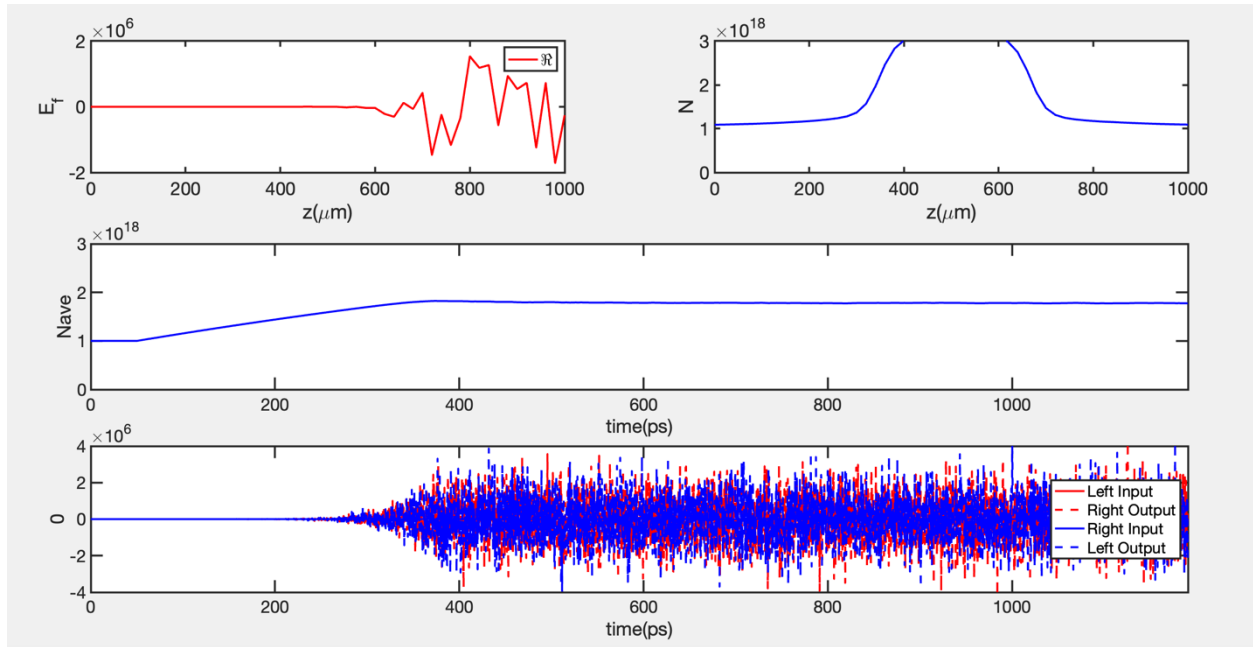


Figure 15: The output plot for the time domain for the wave propagation of the optical signal and carrier density with SPE and no gain polarization

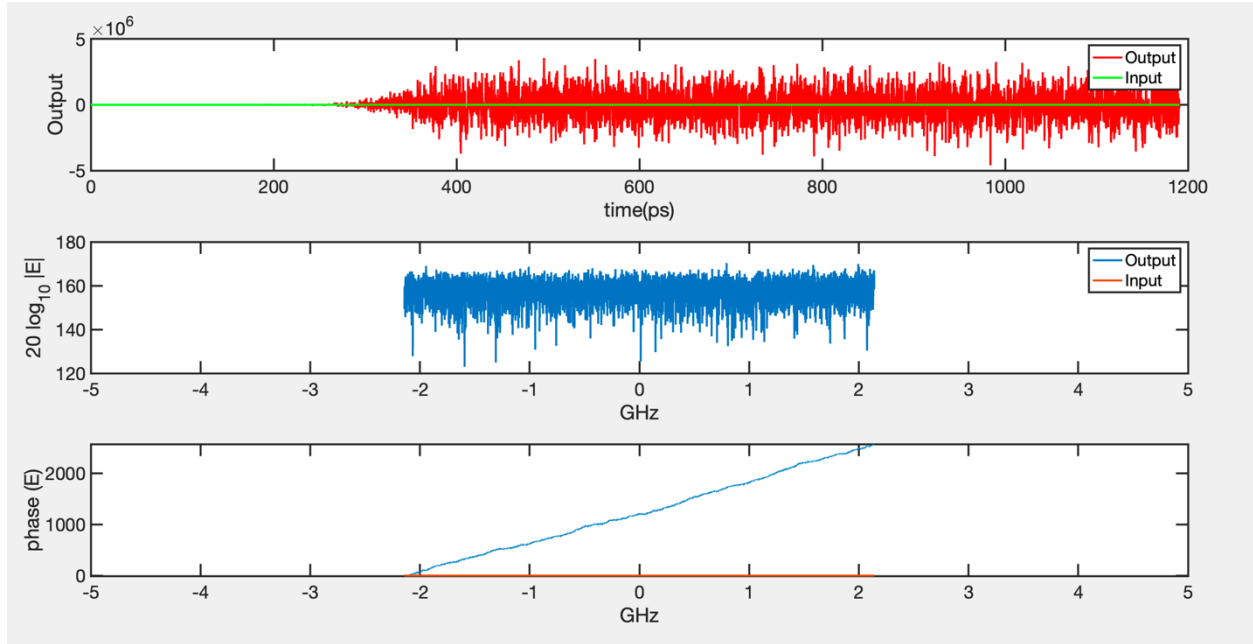


Figure 16: The output plot for the time domain for the wave propagation of the optical signal and carrier density with SPE and no gain polarization

When the system is simulated, a localized electrical pump is applied in the cavity's centre, and spontaneous emission noise is used as the sole optical input. The spatial and temporal dynamics are depicted in Figure 15, which demonstrates that spontaneous emission causes random oscillations in the forward electric field (E_f), which otherwise stays relatively low. With a noticeable peak at the cavity centre, the carrier density profile attests to efficient localized pumping. Long-term carrier injection in the pumped zone is indicated by the average carrier density rising and stabilizing at a high level.

The spectral and temporal output is shown in Figure 16. The signal exhibits the random variations that are typical of noise-driven spontaneous emission in the time domain. The broad spectral distribution of amplified spontaneous emission is reflected in the frequency-domain plot, which displays a broadband output. As would be predicted in the absence of any coherent input signal, the phase response is likewise erratic. Together, these findings show how a gain medium dominated by localized spontaneous emission behaves.

The project architecture states that the pace at which the carrier density is restored in the gain medium is directly impacted by changes in I_D . The carrier population is anticipated to rise with an increase in injection current, leading to higher signal amplification and possibly wider spectrum output. Similarly, the interaction between the optical field and the carrier reservoir can be modulated by changing the input pulse configuration, such as the repetition rate or pulse width. High-energy or tightly spaced pulses, for instance, would cause more noticeable carrier depletion in between pulses, which would show up in both the frequency and temporal domain responses.

Theoretical insights from the course indicate that raising I_D in conjunction with an improved pulse design would improve gain performance and output power. This would probably result in

more noticeable modulation effects in the optical amplifier's spectrum and temporal properties. Finally, some of the effect of ID and the input pulse configuration can be seen in the figures below.

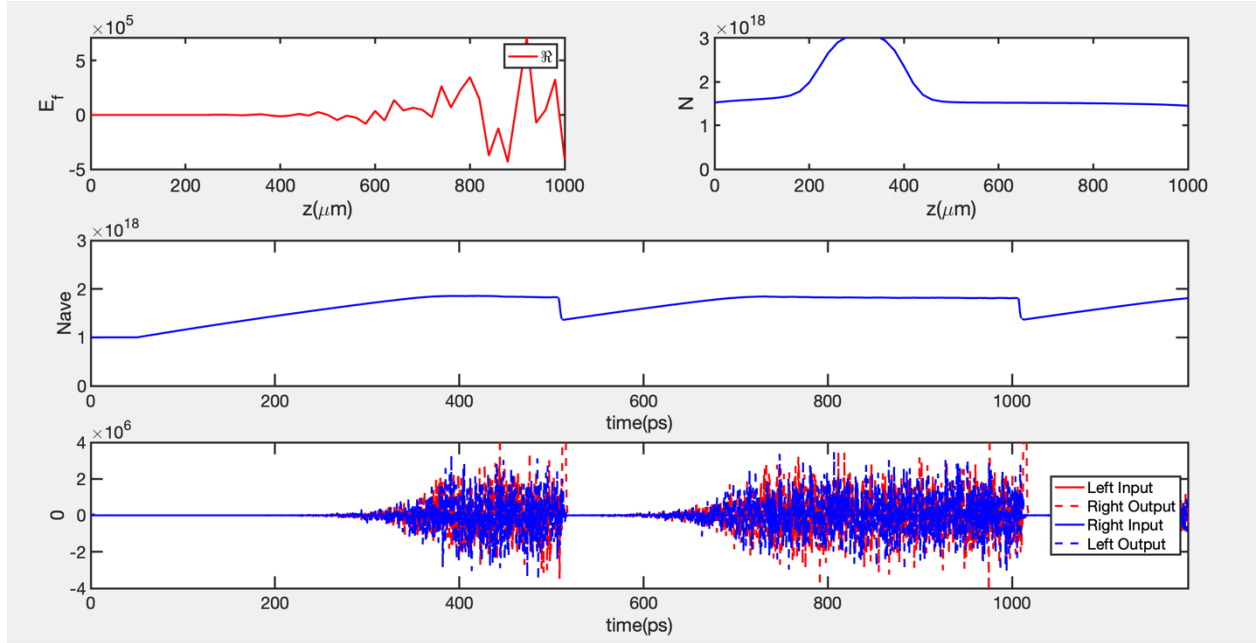


Figure 17: The plot of time domain of wave propagation of the optical signal and carrier density with SPE turned “ON” at a value of 7 and gain polarization is turned “ON” at a value of 0.015

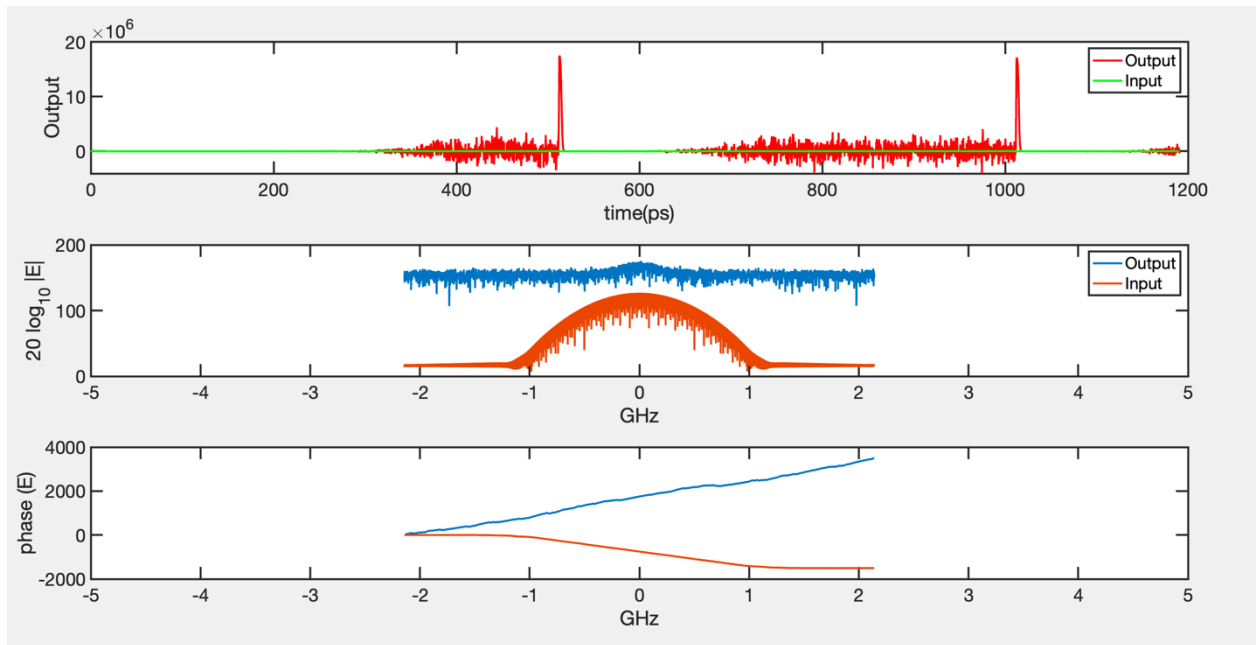


Figure 18: The plot of frequency domain of the wave propagation of the optical signal and carrier density with SPE turned “ON” at a value of 7 and gain polarization is turned “ON” at a value of 0.015

Furthermore, the reason behind the output plot that is seen in the Figures above is due to the change in the value of the code snippet seen below. The difference between the two code snippets can be seen below and what is changed is visible to produce the new Figures (Figure 17 and 18) above.

```
% Input pulse parameters (Modul
InputParasL.E0= 1e5; % Amplitu
InputParasL.we= 0; % Angular
InputParasL.t0= 30e-13; % Time
InputParasL.wg= 10e-13; % Gaus
InputParasL.phi= 0; % phase
InputParasL.rep = 0.5e-9;
InputParasR = 0;
```

Code Snippet 7: The parameters that result in plots shown in Figures 17 and 18

```
% Input pulse parameters (Modulated Gau
InputParasL.E0= 0*1e-5; % Amplitude
InputParasL.we= 0; % Angular Frequen
InputParasL.t0= 30e-13; % Time offset
InputParasL.wg= 10e-13; % Gaussian wid
InputParasL.phi= 0; % phase
InputParasL.rep = 0.5e-9;
InputParasR = 0;
```

Code Snippet 8: The parameters that result in plots shown in Figures 15 and 16

Finally, milestone 7 illustrates the TWM was extended to include both the stimulated and spontaneous emission, enabling a realistic simulation of optical amplifiers and laying the foundation for laser modelling in the next milestone (milestone 8).

The inclusion of the stimulated emission term modelled through the complex gain coefficient allowed the forward and reverse field to experience dynamic amplification based on the local carrier density, introducing the necessary nonlinear interaction between both the optical field and the gain medium.

Furthermore, the SPE (Spontaneous Emission) was implemented using the complex Gaussian white noise to both the forward and reverse electric field and was controlled by **SPE** (with a value of 7) which determined the strength of the SPE noise and **beta_spe** (of value 0.3e-5) which is a scaling factor that influences the magnitude of the noise injected into the fields.

The output's spectral and temporal behaviour is greatly impacted by these variables. For instance, when $SPE = 0$, the system just amplifies the input pulse; but, when SPE is non-zero, the amplifier can also amplify noise, which is crucial when modelling laser startup behaviour.

Additionally, the gain polarization effects-driven by parameters such as:

- Gain: overall gain coefficient
- LGamma: confinement factor
- LGain: length over which the gain is applied

These parameters were already embedded and implemented in the updated field equations which can be found in the MATLAB implementations. These parameters also affect how much gain is applied spatially and how it couples into the mode. Modifying them allows changes to both the

amplitude and spectral content of the output fields, influencing the pulse shape, bandwidth, and frequency-domain characteristics of the system.

Thus, this milestone effectively illustrated how precise modelling of real-world optical amplifiers is made possible by combining carrier dynamics, field-gain coupling, and noise injection. The simulation can now replicate important amplifier behaviours, such as spectrum broadening, noise amplification, and gain saturation, which are essential building blocks for the laser simulations that come next in Milestone 8.

4 Lasers – Milestone 8

By converting the optical amplifier model from Milestone 7 into a laser system, Milestone 8 expands upon the earlier milestones. An active optical device called a laser (Light Amplification by Stimulated Emission of Radiation) uses stimulated emission inside a resonant cavity to produce coherent light. A laser boosts spontaneous emission into a coherent output by maintaining oscillation through feedback from mirrors or gratings, in contrast to an optical amplifier that amplifies an external input signal. The crucial components of laser physics are introduced in this milestone: the interaction between carrier and photon densities, optical feedback through mirrors, and the function of spontaneous emission as the first seed for lasing.

Milestone 8's goals include simulating the laser's turn-on transient, assessing the effect of spontaneous emission (SPE) randomness, using Fourier analysis to examine steady-state (SS) and transient modes, and investigating the effects of injection current (ID) and cavity length (L) on laser behaviour. The understanding of semiconductor lasers used in photonics and telecommunications is based on these tasks, which shed light on laser dynamics such as output stability, mode competition, and threshold conditions.

Theoretical Background

The forward (E_f) and backward (E_r) electric field propagation along a waveguide is described by the Travelling Wave Model (TWM), which is employed throughout this project. In Milestone 7, we modelled an optical amplifier using spontaneous emission and a carrier equation. We now add optical feedback to the laser by adding partially reflecting mirrors with reflection coefficients RL and RR at the ends of the waveguide. Once the gain surpasses the losses, stimulated emission can take over and maintain oscillation because this feedback traps photons inside the cavity.

The population inversion required for gain is provided by the carrier density ($N(z)$), which is decreased by both spontaneous and induced emission and driven by an injection current (I_D). While spontaneous emission (F_f) adds random noise that initiates the lasing process, the photon density ($S(z)$) links to the electric field intensity. The laser's spectrum output is shaped by a frequency-dependent response introduced by the gain polarization, which is represented by the polarization fields (P_f , P_r).

The Key Physical principles include:

- **Threshold Condition:** Lasing happens when the round-trip gain, as measured by internal losses (α) and mirror reflectivities, equals the round-trip loss. As a stochastic source, spontaneous emission starts lasing and affects the distribution of modes.
- **Gain Saturation:** The output stabilizes at steady state as the photon density rises and depletes the carrier density.
- **Cavity Modes:** Observable in the frequency domain by FFT, longitudinal modes are defined by the cavity length (L) and group velocity (v_g).

The governing equations for the laser build on Milestone 7, with modifications for feedback. Deriving them step by step is highlighted below:

Field Propagation Equations:

The TWM equations from Milestone 7, including stimulated and spontaneous emission, are:

$$\frac{1}{v_g} \frac{\partial \hat{E}_f}{\partial t} = -\frac{\partial \hat{E}_f}{\partial z} - i\hat{\beta}(N, S)\hat{E}_f + \hat{F}_f$$

$$\frac{1}{v_g} \frac{\partial \hat{E}_r}{\partial t} = +\frac{\partial \hat{E}_r}{\partial z} - i\hat{\beta}(N, S)\hat{E}_r + \hat{F}_r$$

Here, $\hat{\beta}(N) = \beta_r + i\beta_i$ is the complex propagation constant, with:

$$\beta_i = \frac{1}{2} \left(\frac{g_f G_0 (N - N_{tr})}{v_g} - \alpha \right)$$

- $g_f G_0 (N - N_{tr})$: Gain term, proportional to carrier density above transparency (N_{tr}).
- α : Internal loss (set to 0 in your code).
- $\hat{F}_f, \hat{F}_r$: Spontaneous emission noise terms.

To discretize these using the upwind finite difference method with synchronization ($\Delta z = v_g \Delta t$), we get:

$$\hat{E}_f(t_j, z_i) = f_{\text{sync}} \exp(-i\Delta z \hat{\beta}_{i-1}) \hat{E}_f(t_{j-1}, z_{i-1}) + \hat{F}_f(t_{j-1}, z_i)$$

$$\hat{E}_r(t_j, z_i) = f_{\text{sync}} \exp(-i\Delta z \hat{\beta}_{i+1}) \hat{E}_r(t_{j-1}, z_{i+1}) + \hat{F}_r(t_{j-1}, z_i)$$

where $f_{\text{sync}} = \Delta t v_g / \Delta z = 1$, and $\exp(-i\Delta z \hat{\beta})$ accounts for gain and phase shift.

Figure 19: The field propagation equations and its derivation for the proper implementation needed for milestone 8

Mirror Boundary Conditions:

To form a laser cavity, we introduce mirrors at $z = 0$ and $z = L$:

$$\hat{E}_f(t, 0) = \text{InputL}(t) + RL \cdot \hat{E}_r(t, 0)$$

$$\hat{E}_r(t, L) = RR \cdot \hat{E}_f(t, L)$$

The outputs are:

$$\text{OutputR}(t) = \hat{E}_f(t, L)(1 - RR)$$

$$\text{OutputL}(t) = \hat{E}_r(t, 0)(1 - RL)$$

Setting $RL = RR = 0.5$ (as suggested) provides moderate feedback, balancing gain and output coupling.

Figure 20: The equations for mirror boundary conditions and its outputs

Carrier Equation:

The carrier density evolves as:

$$\frac{dN(z)}{dt} = \frac{\eta I_D}{qV_i} - G_0(N - N_{tr})S(z) - \frac{N}{\tau_n}$$

- $\frac{\eta I_D}{qV_i}$: Carrier injection rate ($\eta = 1$ assumed).
- $G_0(N - N_{tr})S(z)$: Stimulated emission, with $S(z) = (|\hat{E}_f|^2 + |\hat{E}_r|^2)E_{\text{toP}} \cdot 10^{-6}$.
- $\frac{N}{\tau_n}$: Spontaneous recombination.

Discretizing with the forward Euler method:

$$N(t_j, z_i) = \frac{N(t_{j-1}, z_i) + \Delta t \left(\frac{I_{\text{inj}}}{eV_{\text{ol}}} - G_0(N - N_{tr})S \right)}{1 + \Delta t / \tau_n}$$

where $I_{\text{inj}} = I_{\text{on}}$ or I_{off} based on time.

Figure 21: The carrier equation and how it is discretized using the forward Euler method

Spontaneous Emission (SPE):

The noise terms are:

$$\hat{F}_f = eT_f \cdot |\text{SPE}| \cdot \sqrt{N \cdot 10^6}, \quad \hat{F}_r = eT_r \cdot |\text{SPE}| \cdot \sqrt{N \cdot 10^6}$$

where $eT_f, eT_r = A(\text{randn} + i\text{randn})$ if $\text{SPE} > 0$, else $eT_f, eT_r = A$, and:

$$A = \sqrt{\frac{\gamma\beta_{\text{spe}}\hbar f_0 L \cdot 10^{-2}}{2N_z\tau_n}}$$

Polarization Equations:

The gain polarization is:

$$\frac{d\hat{P}_f}{dt} = i\omega'_0\hat{P}_f + \Gamma(\hat{E}_f - \hat{P}_f)$$

Using the trapezoidal rule:

$$\hat{P}_f(t_j) = \frac{\hat{P}_f(t_{j-1}) + 0.5\Delta t \left[\Gamma\hat{E}_f(t_{j-1}) + C_{w0}\hat{P}_f(t_{j-1}) + \Gamma\hat{E}_f(t_j) \right]}{1 - 0.5\Delta t C_{w0}}$$

where $C_{w0} = -\Gamma + i\omega'_0$, and similarly for $\hat{P}_r$.

Figure 22: The spontaneous emission and polarization equation (which helps to acquire the gain polarization)

The main improvement in the MATLAB code for Milestone 8 is the addition of optical feedback, which turns the optical amplifier into a laser. This code builds directly on the framework created in Milestone 7. Partial mirrors are positioned at opposite ends of the cavity to accomplish this. This is accomplished in the simulation by partially reflecting the forward and backward propagating fields by setting the reflection coefficients, RR and RL, to 0.5. A resonant cavity is created by these reflections, and this cavity is necessary for the accumulation of laser oscillations.

Similar to Milestone 7, spontaneous emission noise is still present and is essential for starting a laser. The forward (E_f) and backward (E_r) fields are continuously injected with complex Gaussian noise, which acts as the first seed that, when paired with feedback, initiates lasing.

The dynamics of the carrier density, which are controlled by the interaction of injection current, stimulated emission, and carrier recombination, nevertheless adhere to the previously established rules. To maintain population inversion throughout the simulation, a large injection current is used. The carrier density exhibits a distinctive decrease as the optical field inside the cavity starts to intensify because of an increase in stimulated emission.

With the additional complexity of boundary reflections, the fields are propagated using the same time-stepping approach as in Milestone 7. Constructive interference causes certain longitudinal modes to be selectively amplified as light reflects and travels through the cavity; these modes can be seen in the frequency domain via FFT analysis.

In summary, the modifications implemented in Milestone 8 effectively transition the system from an optical amplifier to a functioning laser. This milestone successfully captures the fundamental behaviors of laser start-up, mode selection, and steady-state operation, highlighting the critical role of feedback and spontaneous emission in laser dynamics.

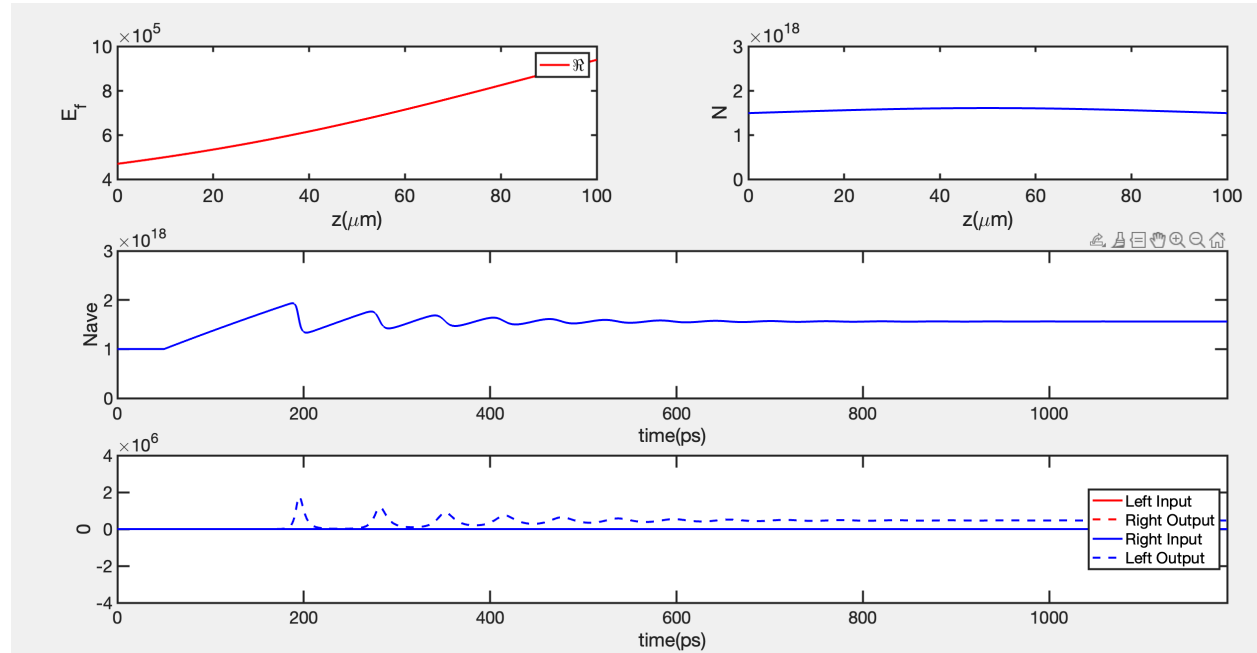


Figure 23: The time domain of the wave propagation in form of a laser

The simulation's main goal was to see the effect of adding an optical injection pulse to the system which affected the dynamic behaviour of the laser cavity. Shedding light or highlighting on how the optical injection alters the cavity's carrier density, electric field, and output signals, the time-domain and frequency-domain behaviours of the system were examined using the MATLAB code that was created.

Looking at the time-domain data in Figure 22, we can see how the output signal, carrier density, and forward electric field change over time. The injected pulse clearly interacts with the electric field plot, as evidenced by oscillations that reveal coupling into the cavity. Electrical injection causes the carrier density to increase initially, but the presence of the optical input dynamically modulates it, exhibiting distinct depletions that correspond to the influence of the pulse. The output signal plot clearly shows the injected pulse, which is followed by several oscillations as the system reacts to the disturbance. As would be expected in an optically injected system, the response of the system is reflected in the left and right output signals, which show signs of field build-up and damping effects.

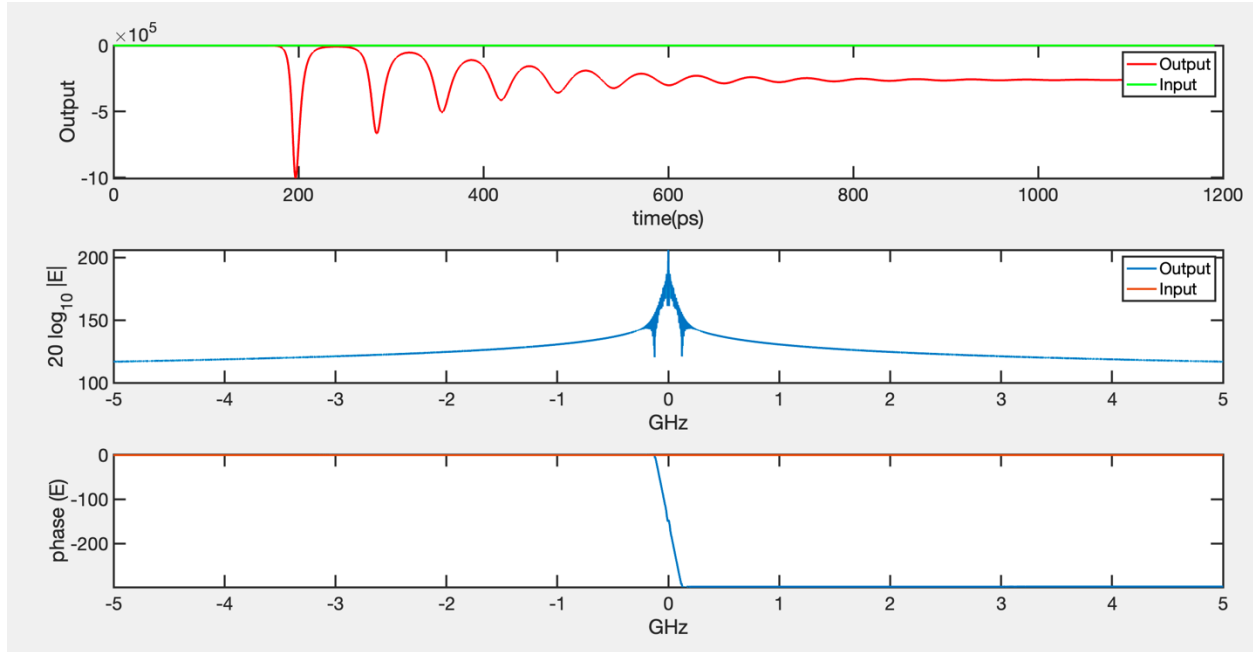


Figure 24: The frequency domain of the wave propagation in form of a laser

The system's reaction to light injection in the frequency-domain results is displayed in Figure 24. Effective coupling of the injected signal into the cavity's resonant modes is shown by the output spectrum's prominent central peak, which is in line with the injection frequency. The modulation effects brought about by the injected pulse are also confirmed by the phase plot, which shows a discernible phase change throughout the frequency range.

The existence of injection locking, in which the injected optical pulse affects and stabilizes the laser cavity's oscillation modes, is strongly supported by these frequency-domain studies. This behaviour is consistent with theory: the carrier dynamics, electric field evolution, and output spectrum content are all changed when an external optical signal is introduced.

Collectively, these results show that optical injection has a major effect on the spectrum and temporal features of the system. The insertion of the optical pulse influences carrier dynamics, electric field evolution, and stabilizes the output frequency components, as predicted theoretically.

The MATLAB implementation of Milestone 8 relies heavily on spontaneous emission noise (SPE) to initiate laser oscillation. The forward (E_f) and backward (E_r) propagating fields are either injected with a deterministic signal or with real random noise, depending on the parameter SPE. The code eliminates the randomness often associated with spontaneous emission by injecting a fixed, constant signal in place of the `randn()` function when SPE is set to a negative value. Although this deterministic seeding can be helpful for debugging, it does not adequately simulate the behaviour of actual lasers, where randomness is necessary to simulate spontaneous photon emission.

This difference is evident when comparing deterministic and stochastic simulations. The use of `randn()` in stochastic runs creates realistic changes in the carrier density and electric fields,

simulating the jitter and noise that are naturally present in real laser systems, particularly during startup. Deterministic runs, on the other hand, lack the random fluctuations that propel natural lasing behaviour and instead display smooth and predictable dynamics. As stressed in the project documentation, this emphasizes the significance of using a randomized technique to simulate the actual properties of spontaneous emission and laser initiation.

Important details about laser operation are also revealed by the frequency-domain study utilizing FFT. The output spectrum is wide during the transient period, suggesting that there are several competing modes. A sharp peak in the FFT output (via `fftshift(fft(OutputR))`) indicates the emergence of a dominant mode as the laser stabilizes and the spectrum narrows. The mode competition and selection that are essential to laser dynamics are shown in this development.

The laser's gain profile and carrier density are directly impacted by changes in the injection current (I_D). Stronger optical output and quicker build-up are the results of higher currents increasing the carrier population. On the other hand, lower injection currents weaken the output signal, decrease gain, and slow the field evolution. The variables `I_on` and `I_off`, which provide the timing and amplitude of the current injection, are used in the code to control this.

Lastly, the longitudinal mode spacing is affected by changing the laser cavity length while keeping the spatial resolution Δz constant. The FFT shows tightly packed spectral lines, which indicate a closer mode spacing caused by a longer cavity. Conversely, a shorter cavity results in a wider mode separation. These findings support the model's correctness in representing the physical connection between cavity length and modal behaviour and are in line with theoretical predictions.

5 Final Example – Milestone 9

By adding cutting-edge visualization techniques, Milestone 9 aimed to expand the capability of the semiconductor laser simulation created in Milestone 8. These improvements were made to shed more light on the laser's dynamic behaviour, especially at startup.

Mode evolution, which visualizes how individual modes increase in strength from noise at startup and reveals the fleeting struggle between modes before the laser settles into its steady-state functioning, was the first enhancement that was decided to be implemented. This was accomplished in the functional MATLAB code, which tracks the evolution of the optical spectrum over time by regularly computing frequency-domain analysis during the simulation. This directly addresses the exploration of time-resolved mode dynamics, one of the open-ended topics recommended in the project outline.

This milestone strengthens the physical understanding of laser dynamics, emphasizing both the transient and steady-state behaviour through spectral evolution tracking.

Visualizing the laser's mode progression during startup is the primary improvement made in Milestone 9. The goal of this addition is to capture the transient dynamics of the laser cavity as spontaneous emission noise seeds the initial field build-up and certain modes gradually take over, building on the steady-state results of Milestone 8. The evolution of the optical spectrum

may be seen by conducting frequency-domain analysis at various points during the simulation. This demonstrates how the initially broad and noisy spectrum becomes sharper when mode competition results in stable lasing. This method offers a more thorough comprehension of the laser's output's temporal evolution, highlighting the crucial change from spontaneous emission to coherent oscillation.

Visualizing the laser's mode progression during startup is the primary improvement made in Milestone 9. The goal of this addition is to capture the transient dynamics of the laser cavity as spontaneous emission noise seeds the initial field build-up and certain modes gradually take over, building on the steady-state results of Milestone 8. The evolution of the optical spectrum may be seen by conducting frequency-domain analysis at various points during the simulation. This demonstrates how the initially broad and noisy spectrum becomes sharper when mode competition results in stable lasing. This method offers a more thorough comprehension of the laser's output's temporal evolution, highlighting the crucial change from spontaneous emission to coherent oscillation.

The main improvement was the addition of specific data structures to hold spectral data as the simulation developed. Mode Evolution and time Vector are two new arrays. The time Vector array tracks the matching simulation times, and the mode Evolution array captures the output signal's power spectral density (PSD) at specific time steps. When combined, these structures allow for time-resolved frequency-domain analysis of the laser output, showing how the system progressively approaches steady-state operation, sharpening the initial broadband noise into separate lasing modes.

The primary simulation loop was changed to effectively capture the spectrum evolution without appreciably lengthening calculation time. $\text{Mod}(i, \text{plotN}) == 0$ was added as a conditional check to sample the output signal on a regular basis. The current output field is extracted at these intervals, and MATLAB's `fftshift(fft())` function is used to calculate its frequency spectrum. The mode Evolution array contains the squared magnitude of the resultant spectrum, while the timeVector array has the matching simulation time. This method guarantees precise tracking of the spectral growth during the brief startup period.

A new figure is created to display the recorded data once the simulation is finished. The stored mode evolution is shown as spectral intensity against frequency and time using MATLAB's `imagesc()` tool. The slow emergence of dominating cavity modes is clearly visible in this picture, providing important information about the dynamics of mode competition and selection during laser startup.

The `InputParasL.E0` parameter was changed to increase the input pulse amplitude, which improved the visibility of spectral characteristics. To seed the laser dynamics and encourage more distinct spectrum growth, this guarantees adequate initial field energy. When taken as a whole, these changes take the simulation beyond steady-state analysis and offer a comprehensive, time-resolved perspective of the dynamic processes controlling laser start-up behaviour.

After implementing this possible design, the following images or figures below are the output.

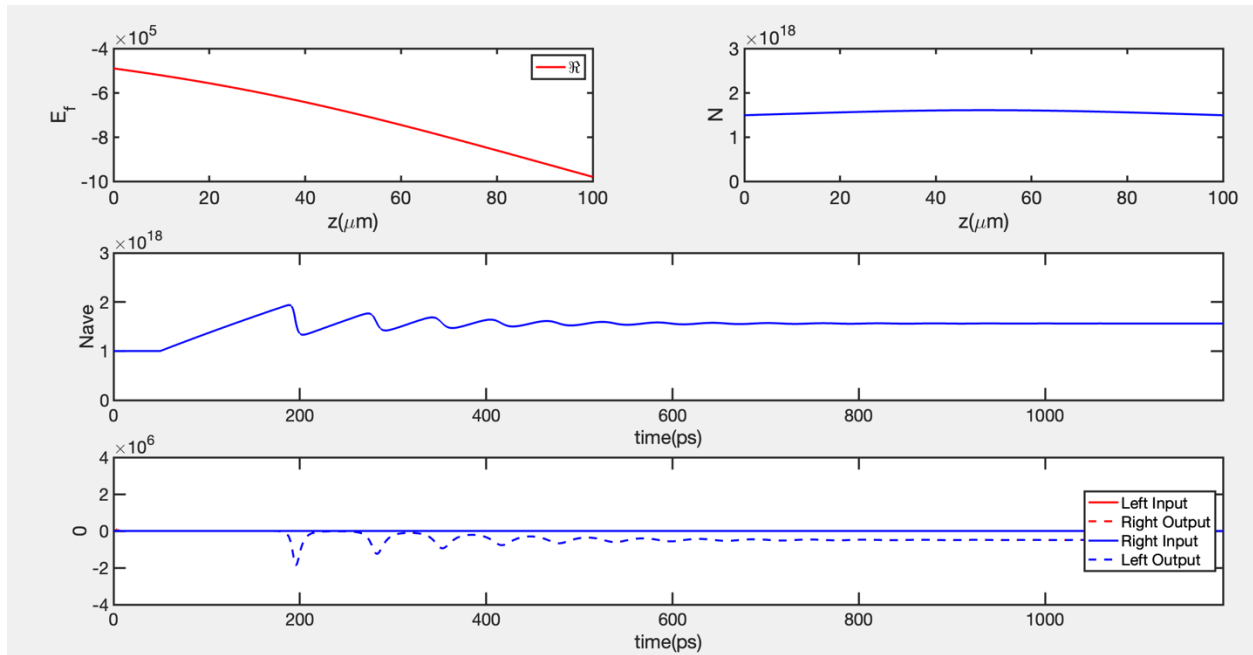


Figure: 25: The time domain output for the wave propagation in the form of a laser

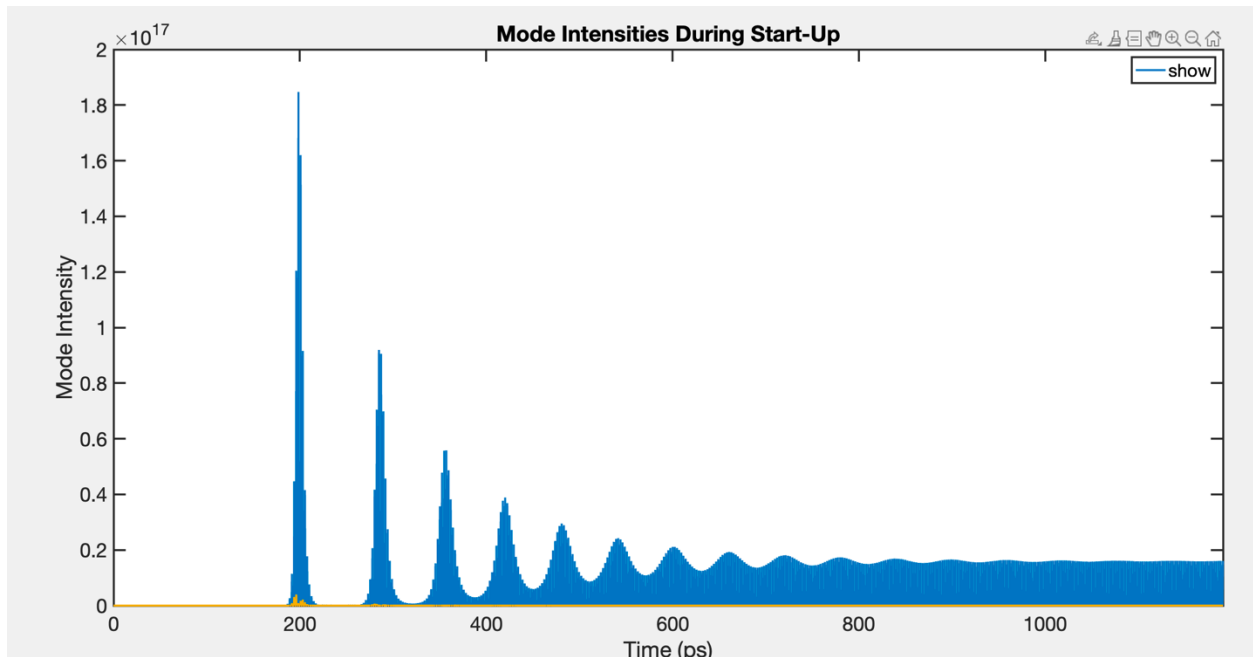


Figure 26: The mode intensities during the start-up of the simulation showing the mode evolution

These findings closely resemble those in Milestone 8, where the forward electric field, carrier density, and optical outputs are depicted in figures. The anticipated increase in field intensity along the cavity is depicted by the forward E_f . The carrier density stabilizes as the population grows.

Field expansion and carrier recombination are confirmed by the output signal's first fluctuations before it settles and becomes stable.

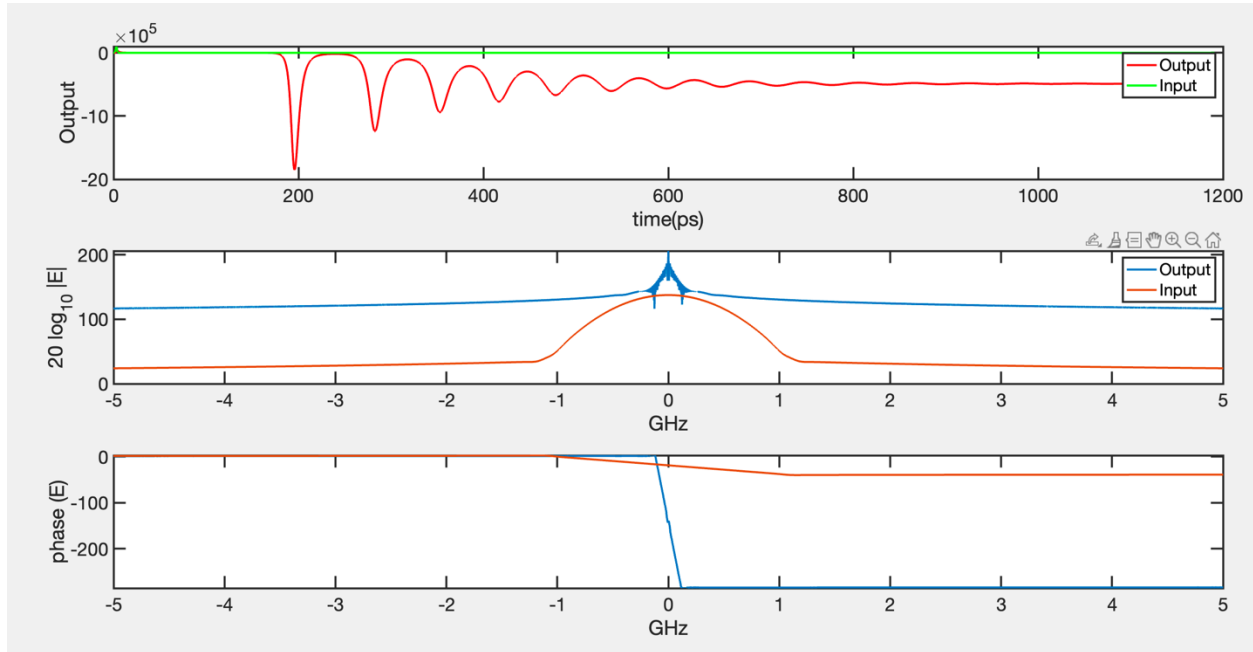


Figure 27: The frequency domain output for the wave propagation in the form of a laser

Additionally, these outcomes are similar to those from Milestone 8. Over time, the oscillations in the time domain signal at the top stabilize. A dominating mode with power build-up is shown at the resonance frequency cavity in the spectral domain plot. Ultimately, a smooth transition is revealed by the phase response, suggesting that coherence has increased among the modes of the system.

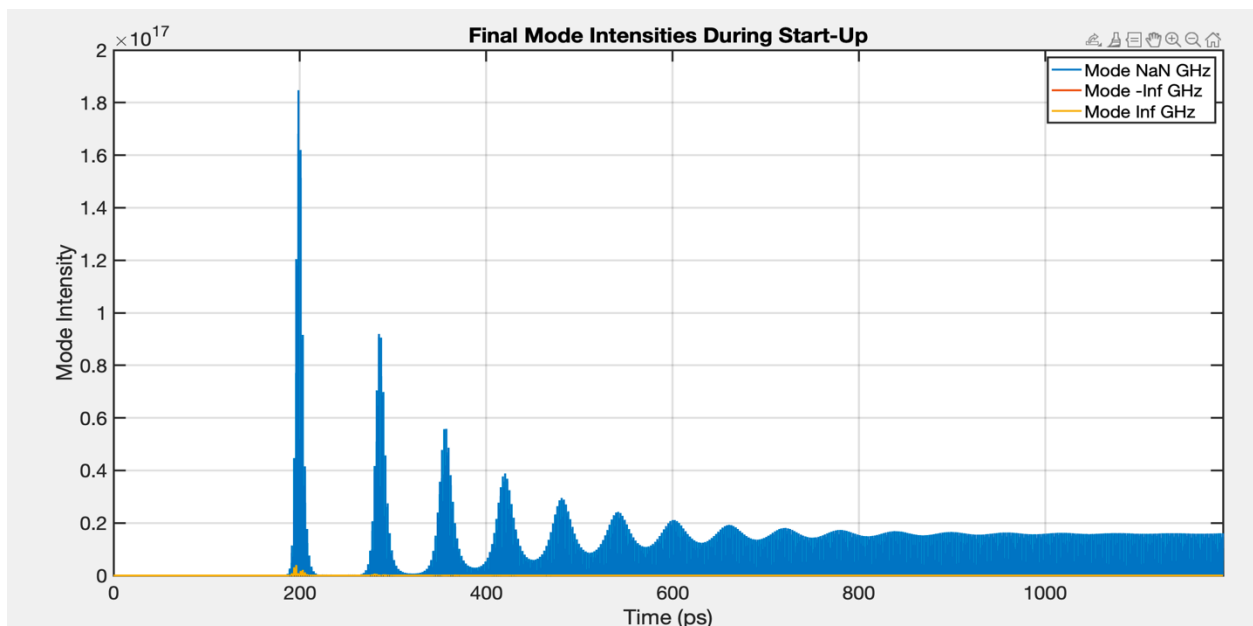


Figure 28: The final mode intensities plot during the startup of the simulation showing the mode evolution

A thorough illustration of the laser's dynamic behaviour throughout startup and steady-state operation can be found in Figures 25 to 28 taken together. Consistent with anticipated laser dynamics, the time-domain results show the steady stabilization of the carrier density and the progressive accumulation of the optical field. The development of spectral coherence and the appearance of dominating lasing modes are further supported by the frequency-domain study.

The mode evolution spectrogram clearly shows the laser spectrum's temporal refinement as it moves from initial broadband noise to a distinct single-mode output. Finally, these visualizations provide a better comprehension of the laser's steady-state behaviour as well as its transient dynamics, meeting the milestone's goals.

6 Conclusion

Through a systematic set of simulation milestones, this research effectively investigated the intricate dynamics of semiconductor lasers. The carrier rate equation's implementation, beginning with Milestone 6, laid the groundwork for modelling the growth of carrier density under diverse electrical and optical excitation scenarios. The basic ideas of carrier dynamics were validated by simulations of both continuous electrical injection and pulsed optical input, which showed important behaviours such carrier buildup, depletion, and the approach to steady state.

In Milestone 7, spontaneous emission noise (SPE) was included to the model to approximate optical amplification. The response of the amplifier to various input signals and noise circumstances was investigated through thorough analysis in the frequency and temporal domains. The simulations highlighted the crucial importance of spontaneous processes in actual amplifier performance by demonstrating how variables like gain polarization, drive current, and SPE intensity affect system stability and output characteristics.

By adding optical feedback through partial mirror reflections, the simulation progressed to Milestone 8, going from an amplifier to a fully functional laser. The system was able to capture the transition from random field build-up to coherent lasing by starting laser oscillation from noise while maintaining the spontaneous emission noise term. Important characteristics that matched theoretical assumptions and offered important insights into laser cavity dynamics were detected, including mode selection, carrier depletion, and spectrum narrowing.

When working on Milestone 9 mode evolution tracking was added as an advanced diagnostic tool to the laser model's capability. A better comprehension of start-up behaviours and steady-state stability was made possible by these improvements, which made it possible to see the emergence of transient modes and field dynamics in the complex plane. The static outputs provided unambiguous insights into the temporal refinement of modes and field coherence, despite technical obstacles to attempts to animate the visualisations.

Numerous difficulties were methodically resolved during the project, including choosing suitable parameter values, guaranteeing numerical stability, and producing understandable data visualizations. Every milestone improved steadily on the one before it, culminating in a solid

simulation framework based on MATLAB that can precisely mimic both transient and steady-state laser events. The project's results demonstrate the effectiveness of computational modelling in assessing and forecasting the behaviour of semiconductor laser systems in addition to reaffirming important theoretical ideas.

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